

Clinical Variant Interpretation Report

Report metadata

Report schema: 1.0 Evidence schema: 1.0 Interpretation prompt: 1.1 LLM model: gpt-5.4-mini Assembly: GRCh38

Case Summary

- Provided HPO terms: HP:0002315
- Matched HPO terms: HP:0002315

Variant Summary

- Assembly: GRCh38
- Variant: MT:5791 G>A
- Population frequency: Not available in the supplied evidence.

Gene and Consequence

- Gene: MT-TC
- Gene ID: ENSG00000210140
- Transcript: ENST00000387405
- Consequence: non_coding_transcript_exon_variant
- Impact: MODIFIER
- Protein change: Not available in the supplied evidence.

Clinical Evidence

ClinVar

- Accession: Not available in the supplied evidence.
- Clinical significance: Not available in the supplied evidence.
- Review status: Not available in the supplied evidence.
- Conditions: Not available in the supplied evidence.

ClinGen

- Disease: mitochondrial disease; Disease ID: MONDO:0044970; Classification: Limited; Mode of inheritance: Mitochondrial

Phenotype Correlation

- Phenotype score: 1
- Provided HPO terms: HP:0002315
- Matched HPO terms: HP:0002315

Interpretation

- The supplied evidence supports that this variant is located in MT-TC and is a non-coding transcript exon variant with MODIFIER impact.

- Gene-disease validity evidence for MT-TC and mitochondrial disease is reported as Limited by ClinGen.
- A ClinVar variant-level classification is not available in the supplied evidence.
- Overall, the supplied evidence is insufficient for a definitive variant interpretation beyond the reported source classifications and limited gene-disease validity evidence.

Limitations

- No ClinVar accession or ClinVar significance was available.
- No population frequency was available.
- No protein change was available.
- The supplied evidence does not include variant-specific clinical assertion from ClinGen; only gene-disease validity for MT-TC was provided.
- The warning states that NCBI ClinVar returned no exact result for this variant.

Evidence source availability

- vep: success
- myvariant: success
- clinvar: not_found
- clingen: success

Report warnings

- NCBI ClinVar returned no exact result for this variant.

References

- Ensembl VEP - https://rest.ensembl.org/vep/homo_sapiens/region
- MyVariant.info - <https://myvariant.info/v1/variant/chrMT%3Ag.5791G%3EA?assembly=hg38>
- ClinGen Gene-Disease Validity via UCSC GenCC - https://search.clinicalgenome.org/kb/gene-validity/CGGV:assertion_c3d92b0e-6753-4355-a982-2f9cf7cad8be-2025-07-14T16:00:00.000Z
- ClinGen - MONDO:0044970 - https://search.clinicalgenome.org/kb/gene-validity/CGGV:assertion_c3d92b0e-6753-4355-a982-2f9cf7cad8be-2025-07-14T16:00:00.000Z
- PubMed - PMID:17724295
- PubMed - PMID:30030363
- PubMed - PMID:34433719
- PubMed - PMID:35252560

Medical Disclaimer

AI-generated decision-support summary based only on the supplied Evidence Object. It is not a diagnosis or treatment recommendation and requires review by a qualified healthcare professional.